

If one were to send a total of 5 million requests per second to the servers, the total number of incoming requests will rise to 5.5 million with 10/11ths of the traffic being useless. If the network equipment at Facebook is configured to drop the requests coming from attacking computers, it would probably still overflow the capacity of the equipment to handle traffic. This might result in packet loss of genuine requests and thus disable the services. And then, not all websites can handle the amount of traffic that Facebook can!

It is important to know why DDoS is one of the biggest weapons that can be used in cyber warfare. The only protection which can be implemented to prevent DDoS from happening is to close the access to the systems from the outer world, i.e. to prevent the access to the system (we are talking about critical systems here) via the Internet and that as such is a very difficult scenario given today's interconnected world.

'Speaking in Gibberish' is important

One of the prime reasons critical networks cannot be run on a separate network is the geographic area and costs involved. Take India for example. The distance between Bangalore and Delhi is about 2000 kilometers. If one critical system were to be located in Delhi and another in Bangalore, more than 2000 KM of networking infrastructure would be needed to interconnect them. Though security is a prime requirement for important computing infrastructure, the costs cannot be neglected. Relying on the Internet not only makes more sense but also saves money.

Now comes the tough part. Communication between any two critical systems when done over the Internet, faces the possibility of interception i.e. data being copied over the wire. This can be done by compromising any of the networking equipment involved in the transfer of data - cryptography comes into the picture (we hope you did go through last month's FastTrack on cryptography).

Though most important data is transferred securely, a mistake can cause the information to leak out into the wild. In addition, weak cryptographic methods can pose a serious threat to the security of sensitive data. The point worth focussing on is the 'weakness' of the algorithm. We already know the security of algorithms depends on two factors: strength of the algorithm and strength of the key. Years of research has yielded quite a number of secure algorithms such as RSA (one of the most famous ones). The part which still remains challengeable is the strength of the key. The only known method to break such algorithms is brute force.